Household archetypes

Table 1. Household profiles, mean (SD)

Cluster	Min. age	Max. age	Members	Education	Households	People
A1	70.2 (11.2)	78.3 (5.0)	1.8 (0.7)	0.5 (0.1)	40.1%	29.9%
A2	21.2 (7.6)	57.2 (8.6)	3.7 (0.9)	0.7 (0.1)	26.9%	40.2%
A3	46.0 (12.8)	63.0 (8.0)	2.3 (0.8)	0.5 (0.1)	22.9%	21.3%
A4	55.6 (17.1)	72.2 (9.0)	2.1 (0.8)	0.8 (0.1)	10.1%	8.6%

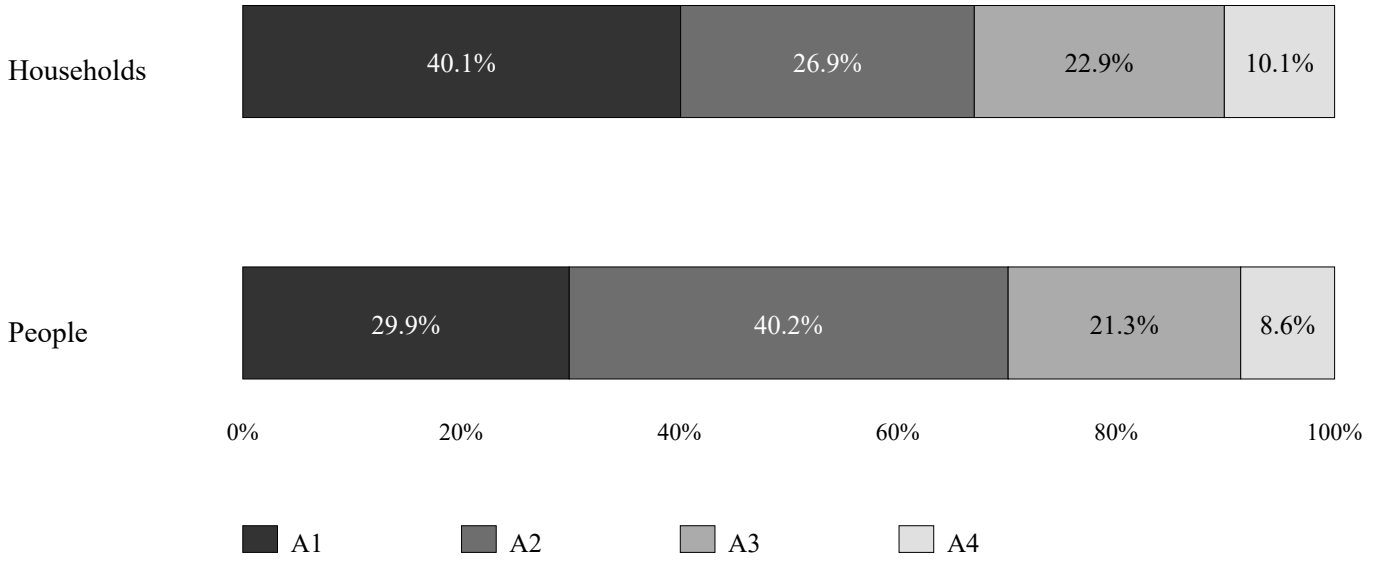


Fig. 1. Archetype shares among households and people.

Source: published Table 3 and Figure 5.

Dwelling area and archetype probability

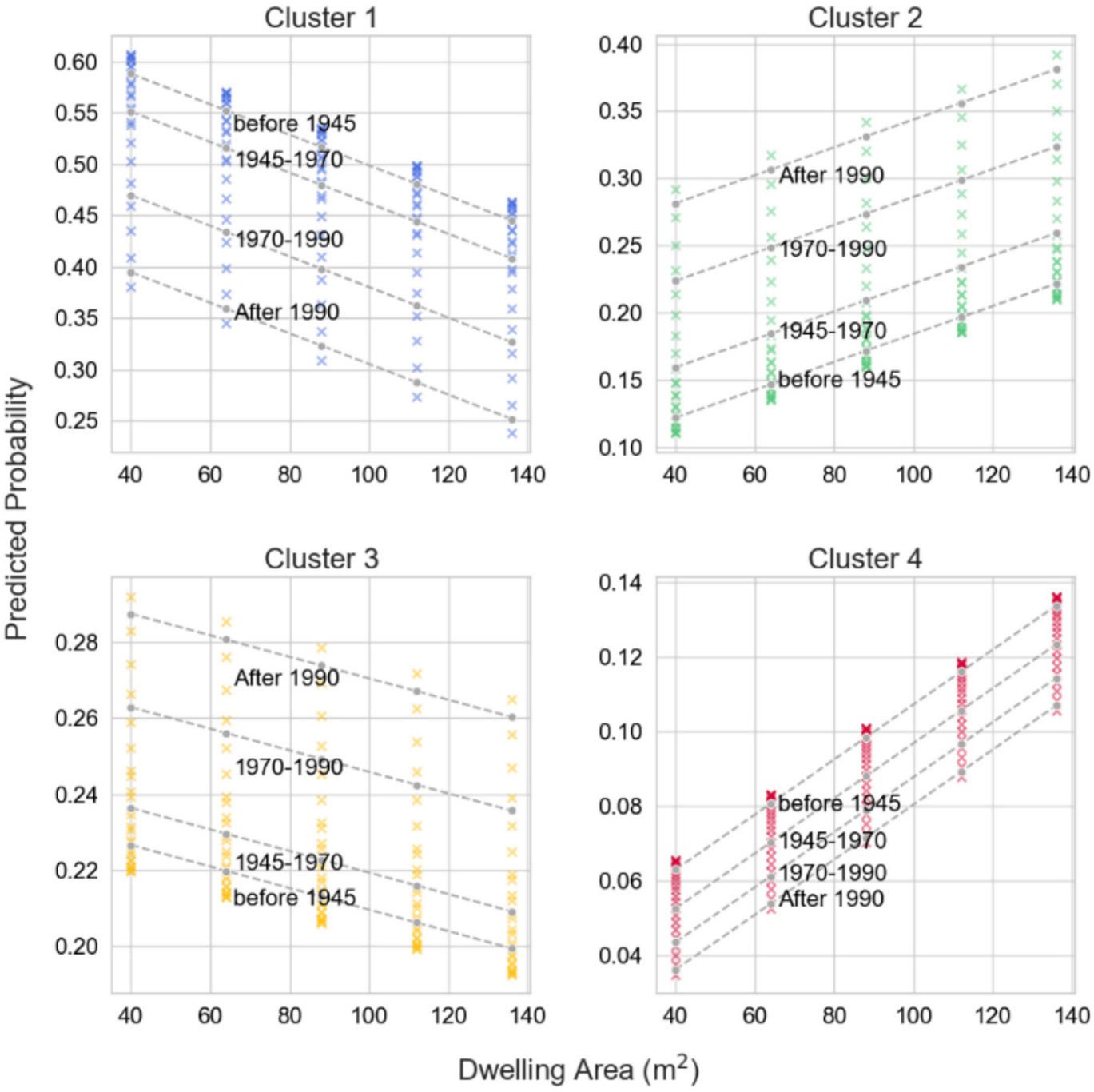


Fig. 2. Archetype probability by dwelling area and construction period.

Source: published Figure 6. Probability scales vary by panel.

Probability model fit

Fitted archetype probabilities

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} 0.289 & 0.692 & 0.000 & 0.000 & -0.372 & -0.084 \\ 0.323 & -0.523 & 0.104 & 0.000 & 0.269 & 0.000 \\ 0.335 & -0.230 & -0.028 & 0.000 & 0.127 & 0.000 \\ 0.033 & 0.039 & 0.000 & 0.014 & 0.152 & 0.000 \end{bmatrix} \times \begin{bmatrix} 1 \\ a / 100 \\ s / 100 \\ as / 10^4 \\ a^2 / 10^4 \\ s^2 / 10^4 \end{bmatrix}$$

a = dwelling age (years); s = floor area (m²).

Coefficients from Eq. (3). Table 4 lists 0.02 for the A4 age² term.

Table 2. Regression fit statistics

Statistic	A1	A2	A3	A4
Adjusted R^2	0.94	0.93	0.87	0.84
RMSE	0.020	0.016	0.008	0.011

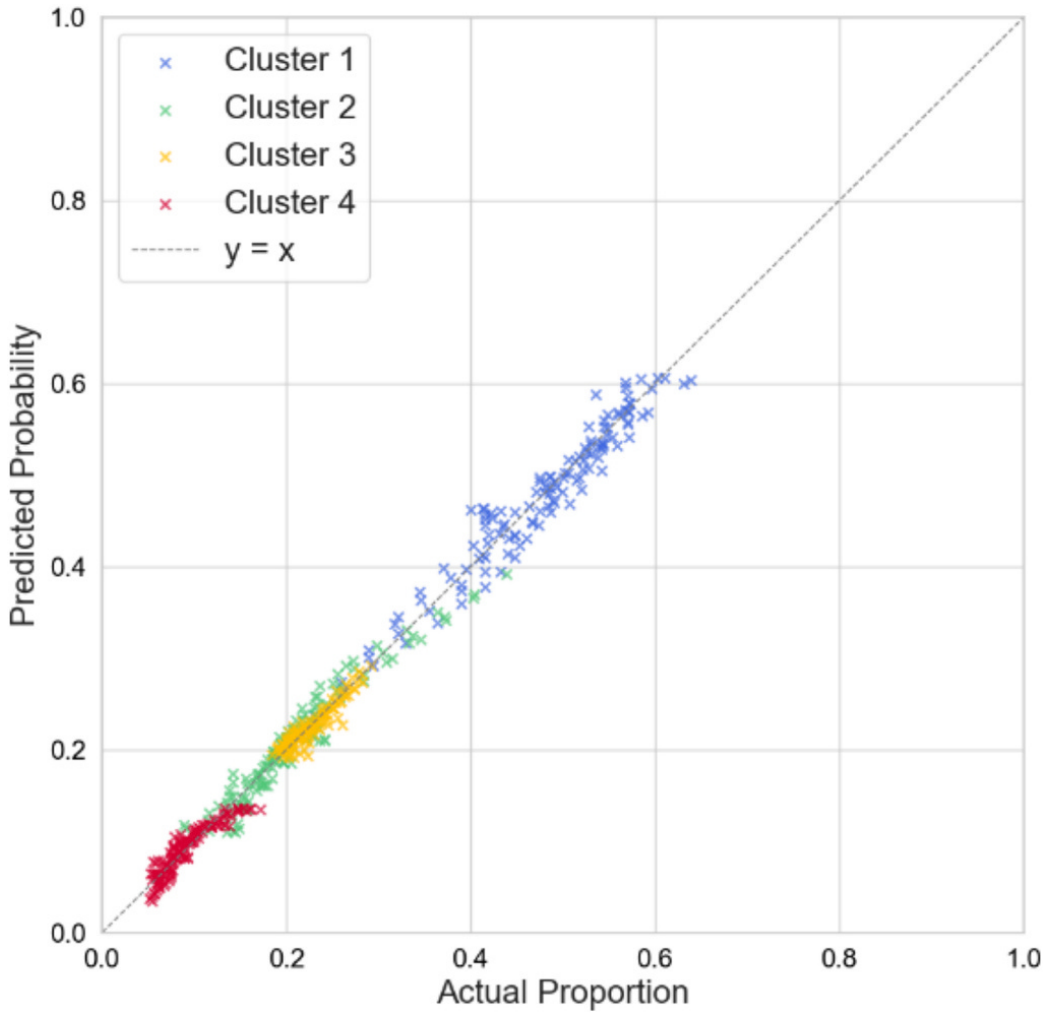


Fig. 3. Predicted probabilities and observed proportions. Source: published Figure 7.

Heating groups

Table 3. Heating system on-time (hours/day)

Group	Mean	SD
H1	4.67	3.54
H2	5.25	3.77
H3	6.91	4.43
H4	7.53	4.41
H5	8.53	4.00
H6	9.14	4.17
H7	9.67	4.28

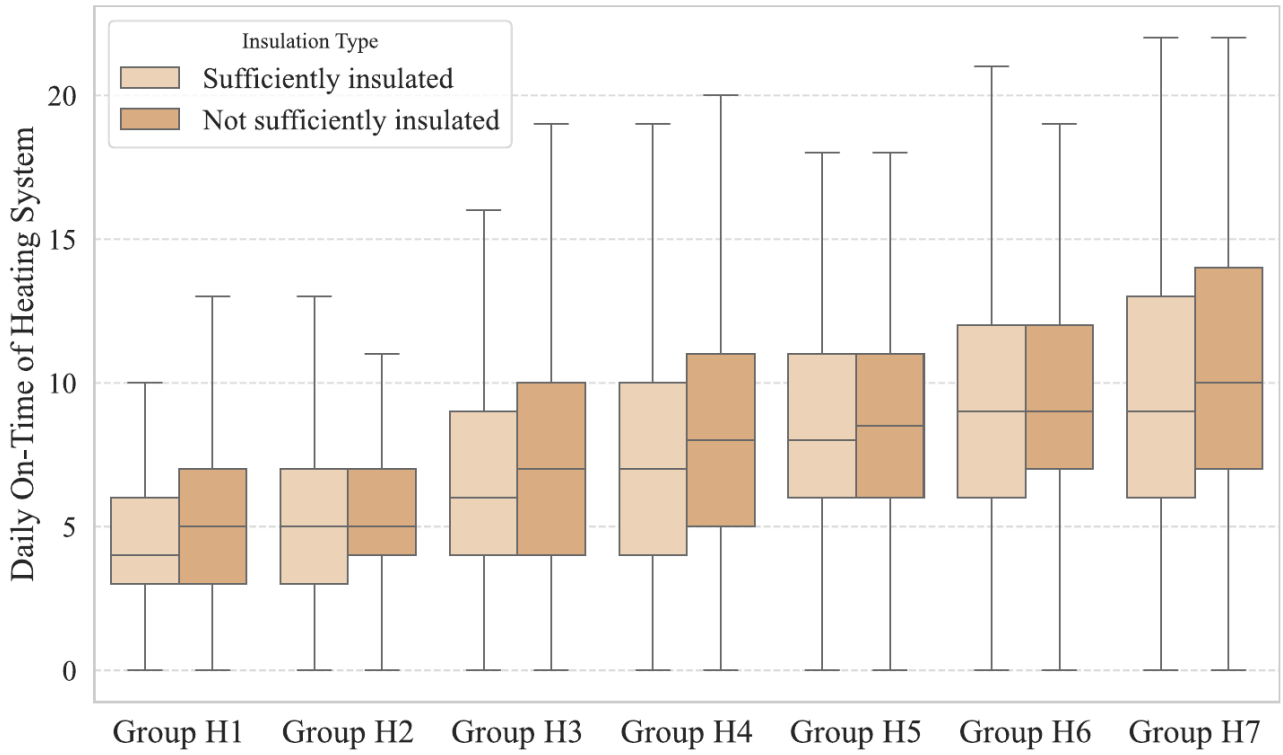


Fig. 4. Heating system on-time by group.

Table 6 and Figure 8(a)

Group labels indicate rank within each year.

Appliance electricity groups

Table 4. Annual appliance consumption (kWh/year)

Group	Mean	SD
E1	1,801	1,502
E2	2,023	1,631
E3	2,320	1,502
E4	2,646	1,844
E5	2,895	1,853

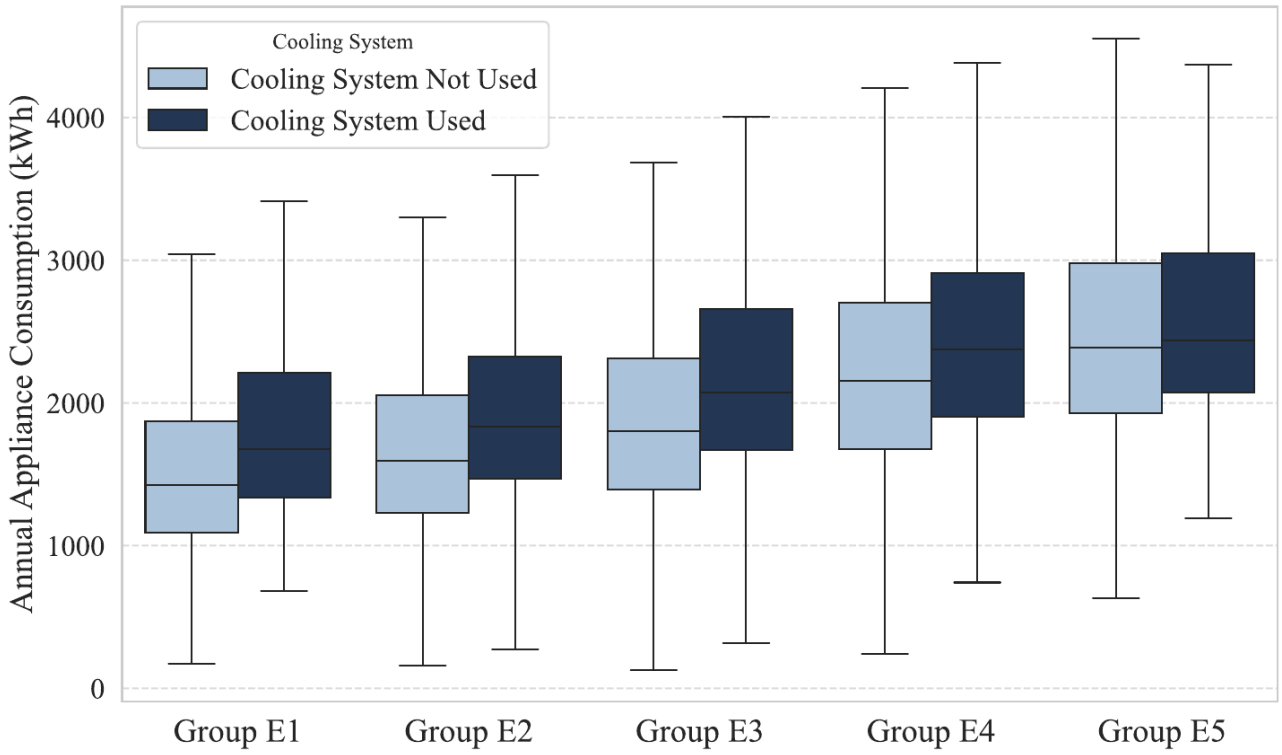


Fig. 5. Annual appliance electricity by group.

Table 6 and Figure 8(b)

Group labels indicate rank within each year.

Appendix A. Group assignments

Table A.1. Energy group by location, dwelling type, and archetype

Location	Dwelling type	Heating				Electricity			
		A1	**A2**	**A3**	**A4**	**A1**	**A2**	**A3**	**A4**
North East	Apartment in small building	H6	H5	H5	H6	E2	E4	E2	E2
	Apartment in medium building	H6	H5	H6	H6	E2	E4	E2	E2
	Apartment in large building	H5	H6	H6	H7	E1	E4	E2	E2
	Single-family house	H7	H6	H6	H6	E2	E5	E3	E4
	Multi-family house	H7	H6	H6	H6	E3	E5	E3	E4
North West	Apartment in small building	H6	H5	H5	H4	E1	E3	E2	E2
	Apartment in medium building	H6	H5	H5	H6	E2	E4	E2	E4
	Apartment in large building	H6	H5	H6	H6	E2	E4	E2	E2
	Single-family house	H7	H6	H7	H7	E2	E5	E4	E4
	Multi-family house	H7	H5	H5	H7	E2	E4	E4	E4
Central	Apartment in small building	H3	H3	H3	H4	E2	E4	E2	E4
	Apartment in medium building	H3	H3	H3	H4	E2	E3	E2	E3
	Apartment in large building	H5	H2	H3	H3	E2	E2	E2	E2
	Single-family house	H4	H4	H3	H3	E3	E4	E3	E4
	Multi-family house	H4	H3	H3	H4	E3	E4	E3	E4
South and Islands	Apartment in small building	H2	H2	H1	H2	E1	E5	E2	E3
	Apartment in medium building	H2	H1	H1	H2	E1	E4	E2	E3
	Apartment in large building	H2	H1	H2	H2	E2	E4	E3	E4
	Single-family house	H3	H4	H3	H3	E1	E5	E3	E4
	Multi-family house	H3	H3	H3	H3	E1	E5	E2	E5

Published Appendix A · apartment classes by building size